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DiffNMR: diffusion models for nuclear magnetic resonance spectra elucidation

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Abstract

Nuclear magnetic resonance (NMR) spectroscopy is a key method for molecular structure elucidation. However, interpreting NMR spectra to deduce molecular structures remains challenging due to the complexity of spectral data and the vastness of the chemical space. Here we introduce DiffNMR, a novel end-to-end framework that leverages a conditional discrete diffusion model for de novo molecular structure elucidation from NMR spectra. DiffNMR refines molecular graphs iteratively through a diffusion-based generative process, ensuring global consistency and mitigating error accumulation inherent in autoregressive methods. The framework integrates a two-stage pretraining strategy that aligns spectral and molecular representations via a diffusion autoencoder and contrastive learning. It also incorporates retrieval initialization and similarity filtering during inference. Our experimental results demonstrate that DiffNMR achieves competitive performance for NMR-based structure elucidation, especially outperforming autoregressive models in domain generalization and robustness, thereby offering an efficient and robust solution for automated molecular analysis.

Supplementary material for this article is available [online](#)

Keywords: NMR structure elucidation, diffusion model, contrastive learning

1. Introduction

Rapid determination of molecular structures may significantly accelerate the process of chemical discovery [1, 2]. Nuclear magnetic resonance (NMR) spectroscopy is one of the most

powerful techniques for determining molecular structures [3, 4]. It provides insights into the local chemical environment of atoms within molecules, revealing details such as functional groups, bond connectivity, and stereochemistry through the magnetic resonance properties of atomic nuclei [5–7]. As a result, NMR has become an indispensable analytical tool in fields such as synthetic chemistry and drug discovery. However, interpreting NMR spectra remains a notoriously challenging task. One major difficulty is spectral complexity, where a single peak may correspond to multiple chemical environments, further complicated by peak overlap and noise interference. Another challenge is the combinatorial explosion of possible structures. As the number of atoms increases, the number of chemically plausible

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structures grows exponentially, making exhaustive or manual searches impractical [8].

NMR interpretation relies heavily on expert knowledge, which is time-consuming and prone to errors. While traditional computer-assisted structure elucidation tools [9, 10] can identify substructures or perform database matching [11, 12], their effectiveness is limited by the small size of reference spectral libraries (which only cover a fraction of known molecules) and the need for manual intervention. With the rapid advancement of automated synthesis and high-throughput experimentation, there is an urgent need to develop end-to-end methods capable of directly inferring molecular structures from NMR spectra without prior knowledge, thereby overcoming critical bottlenecks in the ‘synthesis-analysis’ feedback loop. In recent years, data-driven machine learning (ML) techniques have shown significant potential in NMR spectral analysis [13–15], leveraging the increasing availability of spectral databases and simulated spectra. One class of ML approaches focuses on identifying molecular fragments from patterns in ^1H and ^{13}C NMR spectra and reconstructing complete structures through fragment assembly [16]. This approach emulates chemists’ heuristic reasoning—piecing together a structure from known motifs. Nevertheless, the dependence on predefined fragment libraries and the combinatorial explosion of possible configurations with increasing molecular complexity hinder their applicability to larger systems. Reinforcement learning offers another angle [17–20], where the spectral interpretation task is treated as a sequential decision-making process. A recent example is DeepSPInN [20], which formulates structure elucidation as a Markov decision process and employs a Monte Carlo tree search (MCTS) guided by a neural network policy to iteratively build molecular graphs aligned with given spectra. For small molecules (<10 heavy atoms), it achieves high accuracy by leveraging multimodal spectral (Infrared (IR) and ^{13}C NMR) constraints. However, its practical application is limited by molecular complexity, as the MCTS exploration grows exponentially and becomes a computational bottleneck for larger systems.

Transformer-based [21, 22] deep learning models have recently been applied to NMR-based structure elucidation with notable success [16, 23–25]. Transformers are sequence-to-sequence models adept at capturing long-range dependencies, which makes them well-suited to interpret the complex patterns in spectra and directly output a molecular structure. One such study presented a multitask transformer framework where the model simultaneously predicts substructure sets and complete molecular structure from raw ^1H and ^{13}C NMR spectra [23]. By pretraining the transformer encoder–decoder to map 957 predefined substructures into SMILES strings, the model gains strong prior knowledge of chemical composition and achieves 69.6% Top-15 accuracy for molecules up to 19 heavy atoms. Another related transformer-based approach (LL-NMR) leverages the model’s strengths in natural language processing by converting processed NMR spectral features into a textual sequence [25]. Peak descriptors (chemical shift, multiplicity, intensity, etc) extracted by NMR

processing software (MestReNova) are formatted as tokens. A trained transformer encoder–decoder then translates this ‘NMR-speak’ into the molecule’s SMILES representation, achieving 67% Top-1 accuracy with combined ^1H and ^{13}C NMR data, alongside the molecular formula, as inputs on a million-scale dataset. Despite their success, the autoregressive generation process [26] used by these transformer models has inherent limitations. Its strict, step-by-step generation is fundamentally at odds with the holistic nature of molecular structures and is prone to catastrophic error cascades from even minor initial inaccuracies [27]. Moreover, the lack of robustness becomes a major bottleneck in practice, as experimental NMR spectra are often corrupted by noise. For models [25] that rely on translating discrete peak features into a sequence, this spectral noise can easily introduce errors, leading to a sharp decline in performance. This demonstrates their poor robustness and limited domain generalization when faced with the variability of real-world experimental data.

To address these limitations, we introduce DiffNMR, a novel end-to-end framework for de novo molecular structure elucidation from NMR spectra using a conditional discrete diffusion model [28–32]. In contrast to the sequential, error-prone generation of autoregressive models, DiffNMR constructs the molecular graph through a gradual denoising process. This collective refinement approach inherently enforces global structural consistency at each step, mitigating the risk of catastrophic error cascades. To maximize its effectiveness, our framework leverages a specialized two-stage pretraining strategy, complemented during inference by a retrieval-initialized sampling scheme and similarity-based filtering. Our experimental results demonstrate that DiffNMR not only achieves highly effective structure elucidation but, more importantly, exhibits superior robustness and domain generalization. This superiority is particularly pronounced when confronted with the noisy and variable spectra characteristic of real-world measurements. Thus, DiffNMR stands as a robust and efficient solution poised to advance the frontier of automated molecular analysis.

2. Result

2.1. Overview of DiffNMR

The proposed DiffNMR is an end-to-end framework that leverages NMR spectral data as a strong conditional guide for de novo molecular structure generation. Figure 1 outlines an overview of the diffusion-based NMR elucidation framework, which illustrates its structural components and the processes involved in training and sampling.

The components of DiffNMR. The framework comprises three key components: a molecular encoder, an NMR encoder, and a graph decoder. The molecular encoder, built on a graph transformer architecture [33, 34], extracts feature vectors that encapsulate the structural and chemical properties of molecules. The graph decoder, adapted from the Digress model [29], is a denoising graph diffusion model that consists

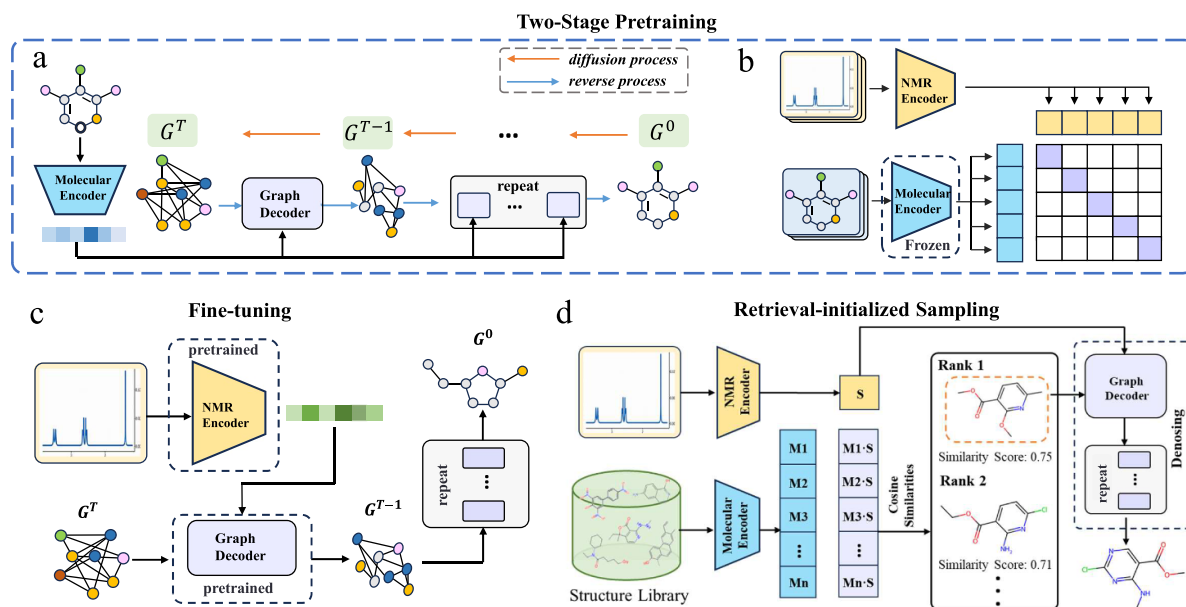


Figure 1. The framework of DiffNMR. (a) and (b) The two-stage pretraining process, in which the first pretraining stage (a) is a molecular diffusion autoencoder (Diff-AE) consisting of a molecular encoder and a graph decoder. The Diff-AE is based on the diffusion model, which defines two Markov processes: diffusion process and reverse process, and is designed to obtain a pretrained graph decoder and a pretrained molecular encoder capable of generating molecular representations. (b) The second pretraining stage—contrastive learning—is designed to align the NMR feature space with molecular representations space by training an NMR encoder with the molecular encoder kept frozen. (c) The fine-tuned end-to-end model, which integrates a pretrained NMR encoder and a pretrained diffusion decoder into a unified system for structure elucidation from NMR spectra. (d) Process of retrieval-initialized sampling. The molecule with the feature vector most similar to the input NMR spectrum is selected from the molecular structure library as the starting point for the denoising process.

of a noise model and a denoising neural network. The noise model follows a Markov process, involving successive graph edits (such as edge addition or deletion, and node or edge category modifications) that can occur independently on each node or edge. The denoising network $\phi_\theta(G^t, t)$ is trained to reverse this process by taking a noisy graph $G^t = (V^t, E^t)$ as input and predicting the clean graph $G^0 = (V^0, E^0)$. It is capable of controllable generation by taking the condition y as an additional input, i.e. $\phi_\theta(G^t, t, y)$. This denoising network employs feature-wise linear modulation [35] in graph transformer layers, where condition-derived parameters adaptively affine-transform hidden features to align the generated graph distribution with y . Additionally, we designed a NMR encoder that preserves the continuous nature of chemical shifts through radial basis function encoding and then integrates the ^1H and ^{13}C NMR modalities via a bidirectional cross-attention mechanism for mutual information enhancement (see details in supplementary information).

Two-stage pretraining strategy. The training of DiffNMR involves a two-stage pretraining strategy. In the first stage (figure 1(a)), a diffusion autoencoder (Diff-AE) [36–39] is trained to establish a mapping between molecular structures and their latent representations. The Diff-AE comprises a molecular encoder, which compresses molecular graphs into compact feature vectors, and a graph diffusion decoder, which reconstructs the original structures through an iterative denoising process. This inherently stochastic

diffusion process leverages randomness to explore a wide range of molecular configurations within the chemical space. This stage yields a pretrained molecular encoder capable of producing compact, informative feature vectors and a pretrained graph diffusion decoder adept at generating molecular structures.

The second pretraining stage aligns NMR spectra with molecular representations in the latent space (figure 1(b)). An NMR encoder is trained to map NMR spectra to the molecular latent space established in the first stage. This alignment is achieved via contrastive learning [40–42], where the NMR encoder minimizes the distance between the latent representations of matching spectrum-molecule pairs while maximizing it for non-matching pairs. The molecular encoder's parameters are frozen during this stage, compelling the NMR encoder to adapt to the pre-established molecular latent space. This process produces an NMR encoder that generates semantically meaningful spectral representations compatible with the diffusion decoder. Furthermore, this alignment enables efficient matching of spectra to molecules in databases, facilitating both retrieval-initialized generation and similarity-based filtering during inference.

Fine-tuning for structure elucidation. Following pretraining, the NMR encoder and graph decoder are unified into an end-to-end model and jointly fine-tuned for direct spectrum-to-structure generation (figure 1(c)). The NMR encoder's output serves as a conditioning signal for the diffusion decoder,

Table 1. Top- k ($k=1, 5, 10$) prediction accuracies for different NMR input modalities. Performance is evaluated with and without the molecular formula as an additional input, with results grouped for molecules containing up to 15, 20, and 25 heavy atoms (HA).

	Formula	≤ 15			≤ 20			≤ 25		
		Top-1	Top-5	Top-10	Top-1	Top-5	Top-10	Top-1	Top-5	Top-10
^1H NMR	×	39.17%	46.69%	51.73%	45.68%	52.53%	56.28%	40.91%	48.40%	52.34%
^{13}C NMR	×	14.61%	22.02%	26.12%	14.56%	21.58%	26.62%	11.60%	18.50%	22.69%
$^1\text{H} + ^{13}\text{C}$ NMR	×	58.83%	66.61%	71.39%	60.34%	67.30%	71.40%	53.10%	61.36%	65.53%
^1H NMR	✓	58.56%	65.54%	69.79%	57.48%	65.09%	69.85%	52.04%	61.04%	65.75%
^{13}C NMR	✓	27.75%	36.34%	42.46%	22.73%	32.92%	40.04%	19.75%	29.94%	37.33%
$^1\text{H} + ^{13}\text{C}$ NMR	✓	68.26%	75.04%	80.27%	67.10%	74.98%	79.58%	61.55%	70.05%	75.07%

guiding the denoising process to generate molecules consistent with the input spectrum.

Retrieval initialization and similarity filtering during inference DiffNMR employs flexible sampling and filtering mechanisms to optimize performance, adapting seamlessly to scenarios with or without an external database (figure 1(d)). When a database is available, DiffNMR utilizes a retrieval-initialized sampling scheme to enhance efficiency and accuracy. By leveraging the contrastively trained encoders, the model retrieves a set of molecules from the database whose latent representations closely align with that of the input NMR spectrum. These retrieved structures serve as informed starting points for the denoising process, reducing the required number of denoising steps and increasing the likelihood of converging to the correct structure. In the absence of a database, generation begins from a random noise vector, a standard approach for diffusion models. This allows the decoder to iteratively denoise and refine the structure, exploring a diverse array of candidate configurations. While this approach broadens chemical space exploration, it may require more iterations to achieve precise results. Regardless of the initialization method, the stochastic nature of the reverse diffusion process yields multiple candidate structures. Contrastive learning provides a robust, quantitative method to select the optimal candidate. By computing the cosine similarity between the latent vector of each generated candidate and that of the input NMR spectrum, DiffNMR ranks and filters the structures. This step discards less relevant candidates and prioritizes those most consistent with the spectrum, thereby ensuring higher precision. The retrieval initialization and similarity-based filtering enhance DiffNMR’s adaptability and robustness across diverse operational scenarios.

2.2. Dataset and evaluation metrics

Our experiments were conducted on the multimodal spectroscopic dataset (MSD), a collection of 794 403 unique molecules from the USPTO reaction dataset [24]. These molecules, composed of C, H, O, N, S, P, and halogens, each contain between 5 and 35 heavy atoms. For each molecule, five complementary spectra were simulated: ^1H NMR ^{13}C

NMR HSQC, IR, and mass spectrometry. All NMR simulations were performed in MestReNova using deuterated chloroform as the solvent under default acquisition parameters. The dataset features detailed spectral annotations. Specifically, for ^1H NMR it provides chemical shifts, multiplet analysis, normalized peak integrations, and coupling constants. For proton-decoupled ^{13}C NMR it includes chemical shifts and relative intensities. This comprehensive spectral information is essential for molecular structure elucidation. Additional details are available in supplementary information. We additionally evaluated our model on NMRexp [43], an open database of 3.37 million experimental NMR records for six nuclei (^1H , ^{13}C , ^{19}F , ^{31}P , ^{29}Si , ^{11}B), extracted from nearly 200k Supporting Information documents published between 2010 and 2024. NMRexp provides rich spectral annotations including chemical shifts, multiplicities, J-couplings, and solvents, and exhibits strong internal reproducibility (MAE ≈ 0.026 ppm for ^1H and ≈ 0.206 ppm for ^{13}C). We conducted out-of-distribution generalization experiments on a subset of this dataset (see table 2), where molecules have ≤ 25 heavy atoms and are composed solely of the element types allowed in the MSD dataset (C, H, O, N, S, P, and halogens).

We assessed model performance with two metrics: Top- k accuracy and Tanimoto similarity. Top- k accuracy quantifies the rate at which the ground-truth structure is found within the top k predictions. Tanimoto similarity measures the structural resemblance between predicted molecules and the ground truth. This similarity is calculated using Morgan fingerprints, generated via the RDKit toolkit [44] with a radius of 2 and a 2048-bit length. A higher Tanimoto score signifies a greater structural likeness.

2.3. DiffNMR enables effective structure elucidation

Our experiments, summarized in table 1, evaluate the model’s performance on molecular structure prediction across datasets of molecules with up to 15, 20, and 25 heavy atoms. The analysis compares the effects of including the molecular formula and using different NMR input types (^1H , ^{13}C , and $^1\text{H} + ^{13}\text{C}$). The results clearly show that prediction accuracy is enhanced when the molecular formula is provided. For the subset with up to 15 heavy atoms, the Top-1 accuracy for the combined

Table 2. Top-1 accuracy of LL-NMR and DiffNMR on the MSD under three noise levels (noise_a, noise_b, noise_c) applied to $^1\text{H}/^{13}\text{C}$ chemical shifts, and on the NMRexp dataset.

			Top-1 accuracy		
			≤ 15	≤ 20	≤ 25
MSD	noise_a	LL-NMR	26.46%	34.94%	32.03%
		DiffNMR	37.15%	39.14%	36.76%
	noise_b	LL-NMR	23.86%	32.16%	28.26%
		DiffNMR	36.49%	38.59%	35.94%
	noise_c	LL-NMR	22.66%	30.15%	25.95%
		DiffNMR	35.50%	36.87%	33.98%
NMRexp		LL-NMR	7.94%	6.30%	5.55%
		DiffNMR	18.07%	12.61%	10.69%

$^1\text{H} + ^{13}\text{C}$ NMR input increases from 58.83% (without formula) to 68.26% (with formula). This trend holds consistently across the datasets with up to 20 and 25 heavy atoms. This enhancement arises because the molecular formula acts as a powerful structural constraint, specifying the elemental composition and thereby drastically reducing the combinatorial search space.

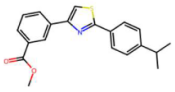
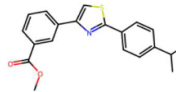
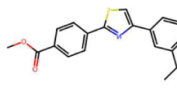
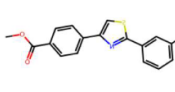
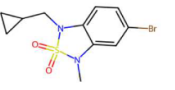
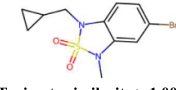
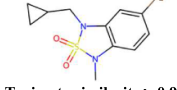
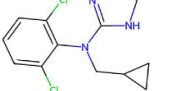
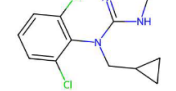
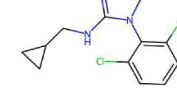
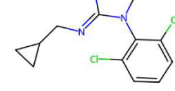
Furthermore, a comparison of input modalities reveals the distinct advantages of ^1H NMR over ^{13}C NMR and the synergistic benefit of combining both. For instance, in the formula-included condition for molecules with ≤ 15 heavy atoms, the Top-1 accuracy of ^1H NMR (58.56%) substantially surpasses that of ^{13}C NMR (27.75%). This is primarily because ^1H NMR provides richer structural details, such as coupling patterns and integration values that describe local hydrogen environments. In contrast, ^{13}C NMR offers fewer signals, though they are crucial for defining the carbon backbone. The highest performance is achieved when ^1H and ^{13}C NMR spectra are used together, reaching a Top-1 accuracy of 68.26%. This synergy occurs because ^{13}C NMR complements ^1H NMR by resolving ambiguities in the carbon skeleton and distinguishing functional groups with overlapping proton signals, together forming a more complete spectral profile. The superior accuracy of the combined input underscores that our NMR encoder effectively fuses information from both spectra. Given this advantage, our subsequent experiments focus exclusively on the $^1\text{H} + ^{13}\text{C}$ NMR input configuration.

Moreover, analyzing accuracy across the datasets highlights an intricate balance between molecular complexity and dataset size. For the combined $^1\text{H} + ^{13}\text{C}$ NMR input (without formula), Top-1 accuracy first increases slightly from 58.83% (≤ 15 HA) to 60.34% (≤ 20 HA), but then drops to 53.10% (≤ 25 HA). The initial rise in accuracy from ≤ 15 HA to ≤ 20 HA can be attributed to the increased information content within slightly larger molecules. As molecules expand in this range, they typically offer more distinct structural features (e.g. a greater diversity of NMR signals and richer chemical environments), which better equip the model to differentiate isomers. Simultaneously, the training dataset significantly grows by approximately 120% (from $\sim 120\,000$ to

$\sim 260\,000$ structures). This ample data expansion appears sufficient to cover the increased structural diversity at 20 HA, allowing the model to effectively leverage the enhanced structural information and sustain, or even modestly improve, its performance. However, the subsequent drop in accuracy from ≤ 20 HA to ≤ 25 HA suggests that beyond a critical complexity threshold, the benefits of enriched structural information are overwhelmed by the escalating difficulty of the task. For molecules exceeding 20 heavy atoms, the number of potential isomers skyrockets exponentially, making their structural distinctions increasingly subtle and challenging for the model to resolve. This exponential surge in structural diversity and task complexity far outpaces the data growth. Although the training data still increases by 50% (from $\sim 260\,000$ to $\sim 390\,000$ structures) in this range, this insufficient expansion fails to keep pace with the dramatically heightened challenge of differentiating a vast, highly similar structural space, ultimately resulting in a 7% accuracy drop.

Beyond evaluating performance on clean, idealized data, a critical test of a model's practical utility is its robustness against the noise inherent in real-world experimental spectra. To assess this, we compared the performance of DiffNMR against LL-NMR [25], a representative autoregressive transformer-based model, on the noisy simulated dataset MSD and the experimental dataset NMRexp. Although other advanced models exist, such as multitask-NMR [23] which processes raw 1D NMR spectra as arrays, we selected LL-NMR as the primary baseline because it shares the same processed peak string input (including chemical shifts, multiplicities, integrals, etc) as DiffNMR, which simplified the input, reduced the learning demand on the model, and focused the model on higher-level reasoning, enabling a fair comparison that isolates the advantages of our diffusion-based paradigm over autoregressive generation. For the MSD, we introduced two types of noise to emulate common experimental imperfections: structural noise, where one ^1H and one ^{13}C peak were randomly added or removed to simulate peak-picking errors, and intensity noise, where uniform noise was added to the $^1\text{H} / ^{13}\text{C}$ chemical shifts at three progressively severe levels: $\pm 0.01 / \pm 0.1$ ppm (noise_a), $\pm 0.02 / \pm 0.2$ ppm

a correctly predicted molecule

target molecule	predicted molecule		
	 Tanimoto similarity: 1.00	 Tanimoto similarity: 0.86	 Tanimoto similarity: 0.83
	 Tanimoto similarity: 1.00	 Tanimoto similarity: 0.99	 Tanimoto similarity: 0.87
	 Tanimoto similarity: 1.00	 Tanimoto similarity: 0.83	 Tanimoto similarity: 0.76

b incorrectly predicted molecule

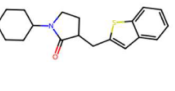
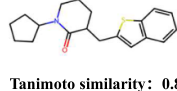
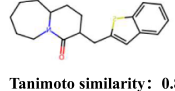
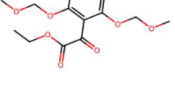
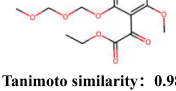
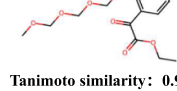
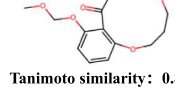
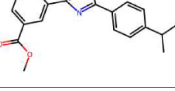
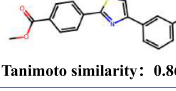
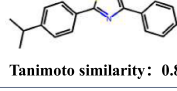
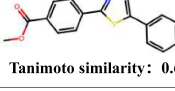
target molecule	predicted molecule		
	 Tanimoto similarity: 0.91	 Tanimoto similarity: 0.86	 Tanimoto similarity: 0.84
	 Tanimoto similarity: 0.98	 Tanimoto similarity: 0.91	 Tanimoto similarity: 0.81
	 Tanimoto similarity: 0.86	 Tanimoto similarity: 0.82	 Tanimoto similarity: 0.60

Figure 2. Examples of Top-3 sampled molecular structures with their Tanimoto similarity scores to the target molecule. (a) Correctly predicted molecules: The first predicted molecule in each set matches the target perfectly (Tanimoto similarity = 1), while the other two predictions have slightly lower similarities. (b) Incorrectly predicted molecules: None of the Top-3 predicted molecules match the target exactly, with Tanimoto similarities less than 1.

(noise_b), and $\pm 0.03/\pm 0.3$ ppm (noise_c). The comparative results, detailed in table 2, demonstrate the superior robustness and domain generalization of DiffNMR. On the MSD, DiffNMR consistently outperforms LL-NMR across all noise levels. For instance, under mild noise (noise_a) for molecules with ≤ 15 heavy atoms, DiffNMR achieves 37.15% Top-1 accuracy, exceeding LL-NMR's 26.46% by over 10%. While both models degrade with increasing noise, DiffNMR shows greater resilience; on the ≤ 25 HA dataset, its accuracy drops only 2.78% from noise_a to noise_c (36.76% to 33.98%), versus LL-NMR's 6.08% decline (32.03% to 25.95%). This highlights DiffNMR's holistic, parallel denoising process, which is more robust to localized errors than LL-NMR's sequential autoregressive approach, vulnerable to error cascades from misinterpreted peaks. On the more complex real-world NMRexp dataset, both models show significant accuracy declines. This drop underscores the challenges of out-of-distribution generalization, where models trained on simulated spectra must adapt to variations in real experimental conditions. Nonetheless, DiffNMR maintains stronger

generalization, e.g. 18.0% Top-1 accuracy for ≤ 15 HA molecules versus LL-NMR's 7.94%, demonstrating stronger generalization capability of our model.

Figure 2(a) provides several illustrative examples of correctly predicted molecules. In both cases, the top-ranked prediction exactly matches the target molecule (Tanimoto similarity = 1.00), demonstrating the model's ability to resolve structural connectivity with high fidelity. Notably, even in cases where the model fails to rank the correct molecule within the top three predictions, as shown in figure 2(b), the predicted structures still demonstrate high Tanimoto similarity scores relative to the target molecules. These scores reflect a significant degree of structural overlap, as the predicted molecules preserve key chemical motifs—such as functional groups and bonding patterns—present in the target structures. This retention of critical features suggests that, even in failure, the model generates outputs that are structurally relevant and can serve as useful starting points for further refinement or manual analysis. Such behavior underscores the model's robustness, offering valuable

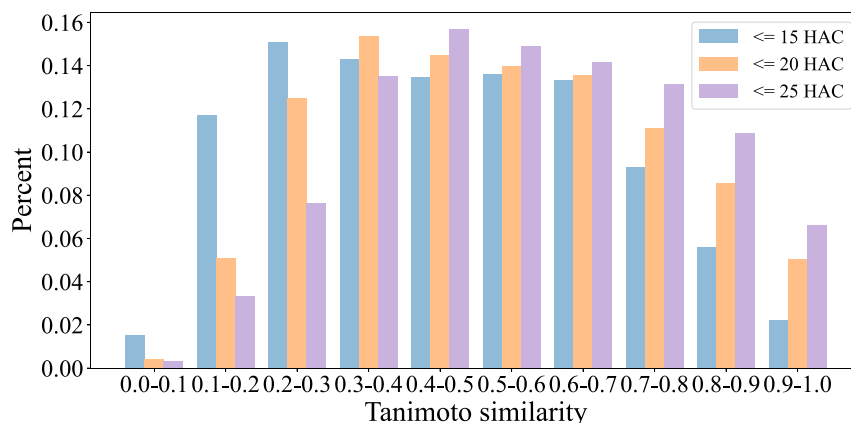


Figure 3. Distribution of Tanimoto similarity scores for incorrect Top-1 predictions (excluding exact matches) across molecules with heavy atom counts ≤ 15 , ≤ 20 , and ≤ 25 . The invalid SMILES strings and the correctly-predicted molecules were excluded.

Table 3. Performance metrics for DiffNMR without molecular formulas, showing Top-1 accuracy and average Tanimoto similarity under varying conditions of similarity-based filtering and retrieval initialization.

Similarity-based filtering	Retrieval initialization	≤ 15		≤ 20		≤ 25	
		Accuracy	Tanimoto similarity	Accuracy	Tanimoto similarity	Accuracy	Tanimoto similarity
×	×	51.11%	0.73	43.25%	0.72	35.76%	0.71
×	√	53.87%	0.75	53.07%	0.78	49.70%	0.78
√	×	58.83%	0.78	60.34%	0.81	53.10%	0.79
√	√	60.13%	0.79	61.18%	0.82	58.47%	0.82

insights into the system under study despite not achieving an exact match.

The distribution of Tanimoto similarities for incorrect top-1 predictions, stratified by heavy atom count (HAC), is presented in figure 3. Strikingly, the fraction of incorrect predictions with a Tanimoto similarity of 0.5 or greater increases with molecular complexity: 44.0% (≤ 15 HAC), 52.2% (≤ 20 HAC), and 59.6% (≤ 25 HAC). Moreover, the entire similarity distribution shifts toward higher values as the HAC increases. This trend can be attributed to two primary factors. First, larger molecules often exhibit more distinct spectral fingerprints due to their extended carbon frameworks and diverse functional groups, enabling the model to more easily capture broad structural patterns. Second, the training data for high-HAC molecules (approx. 390k for ≤ 25 HAC) likely contains a higher prevalence of recurring motifs (e.g. polycyclic systems or branched chains), which the model learns to prioritize during generation. However, the combinatorial explosion of isomerism at higher HACs makes it challenging to resolve fine-grained structural details, leading to minor deviations in substituent placement or stereochemistry. The trends observed in figures 2(b) and 3 have practical implications: while exact structure prediction remains the ultimate goal, the ability to produce high-similarity outputs in failure cases enhances the model's value as an effective tool for chemists.

2.4. Two-stage pretraining and inference strategies enhance structure elucidation

To further enhance the performance of DiffNMR, we evaluated two key inference strategies: similarity-based filtering and retrieval initialization. Both strategies leverage pretrained representations from contrastive learning to refine the structure generation process. Table 3 presents the Top-1 accuracy and average Tanimoto similarity across datasets with different HACs, focusing on the scenario without molecular formulas.

Similarity-based filtering works by computing cosine similarity scores between the feature vectors of generated molecular candidates and the input NMR spectrum, then ranking and filtering them to pinpoint the structure most consistent with the spectral data. This approach yields notable improvements, as shown in table 3. For the ≤ 15 HAC dataset, Top-1 accuracy rises from 51.11% (without filtering) to 58.83% (with filtering). The impact is even more pronounced for larger molecules: in the ≤ 25 HAC dataset, accuracy jumps from 35.76% to 53.10%, and the average Tanimoto similarity increases from 0.71 to 0.79. These consistent gains underscore the ability of similarity-based filtering to effectively select the most spectrally aligned structure from a diverse pool of candidates, thereby boosting prediction reliability.

This filtering capability is underpinned by the contrastive learning pretraining, as demonstrated in figure 4. We

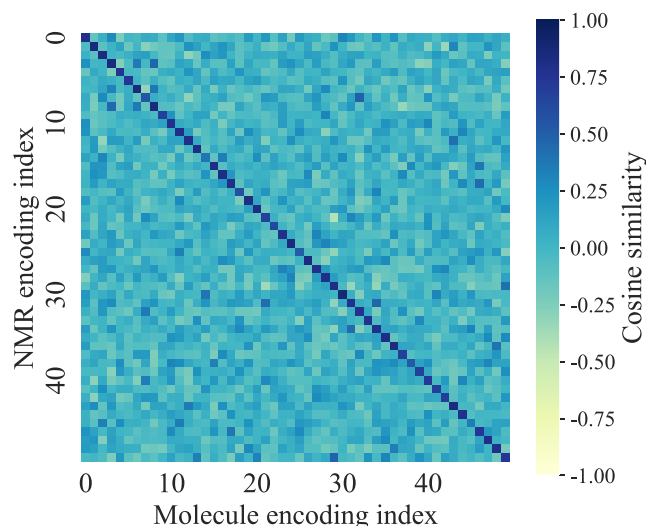


Figure 4. Heatmap showing cosine similarity scores for 50 randomly selected NMR-molecule pairs from the test set. The diagonal corresponds to matching NMR-molecule pairs, and off-diagonal elements represent non-matching pairs.

randomly selected 50 NMR-molecule pairs from the test set and computed the cosine similarity scores between their encoded representations, visualized as a heatmap. The diagonal entries—representing matching NMR-molecule pairs—consistently show the highest similarity scores, while off-diagonal entries (non-matching pairs) exhibit significantly lower values. This pronounced separation confirms that contrastive learning successfully aligns NMR spectra with their corresponding molecular structures in the latent space, providing a robust foundation for similarity-based filtering. By leveraging this alignment, DiffNMR accurately ranks candidates, ensuring the top prediction closely matches the input spectrum.

Additionally, the denoising process benefits from retrieval initialization, which uses a database (in this case, the training set) to provide informed starting points for the diffusion process. This markedly improves prediction accuracy, especially for larger molecules. According to table 3, when filtering is enabled, retrieval initialization increases Top-1 accuracy from 58.83% to 60.13% for the ≤ 15 HAC dataset. This enhancement becomes more substantial as complexity rises: for the ≤ 25 HAC dataset, accuracy improves from 53.10% to 58.47%. Notably, the performance gap between smaller (≤ 15 HAC) and larger (≤ 25 HAC) molecules—previously observed as a 7% decline—narrows to a mere 2% decrease with retrieval initialization. This mitigation suggests that providing chemically relevant initial structures effectively constrains the denoising process, bridging the accuracy disparity caused by increased molecular complexity.

We visualize the molecular structure at different diffusion time steps in figure 5 to illustrate how retrieval initialization leverages prior knowledge from the database to narrow the search space. For a target molecule, we compare two approaches: (i) starting from a random graph, which yields an incorrect prediction after 500 denoising steps, and (ii) using retrieval initialization, which selects

a structurally similar molecule from the database as a starting point, leading to the correct structure in just 200 steps. Initializing with a spectroscopically informed structure enables the model to refine a near-correct candidate rather than exploring blindly. These examples reveal that retrieval initialization not only enhances accuracy but also accelerates sampling efficiency by reducing the required denoising steps. This informed approach is particularly impactful for larger molecules, where random initialization struggles with the vast combinatorial space.

The combined application of similarity-based filtering and retrieval initialization maximizes DiffNMR's performance. In the ≤ 25 HAC dataset, Top-1 accuracy (without formulas) reaches 58.47% compared to 35.76% without either strategy, and average Tanimoto similarity climbs from 0.71 to 0.82. This improvement of performance stems from informed starting points narrowing the search space and similarity-based filtering ensuring precise final selections. Together, these strategies enhance both the precision and structural relevance of the generated molecules.

To assess the effectiveness of the proposed two-stage pre-training strategy in the DiffNMR framework, we conducted experiments on the dataset with ≤ 15 heavy atoms, evaluating the impact of pretraining on Top-1 prediction accuracy. The first-stage pretraining Diff-AE provides pretrained parameters for the graph decoder, while the second-stage pretraining contrastive learning provides pretrained parameters for the NMR encoder. We compared the Top-1 accuracy in four scenarios: no pretraining applied to either component, pretraining applied solely to the graph decoder using Diff-AE, pretraining applied only to the NMR encoder via contrastive learning, and pretraining applied to both components together. These experiments were conducted without retrieval initialization or similarity filtering during the inference stage. The results, depicted in figure 6, show that pretraining either component individually improves the accuracy by approximately

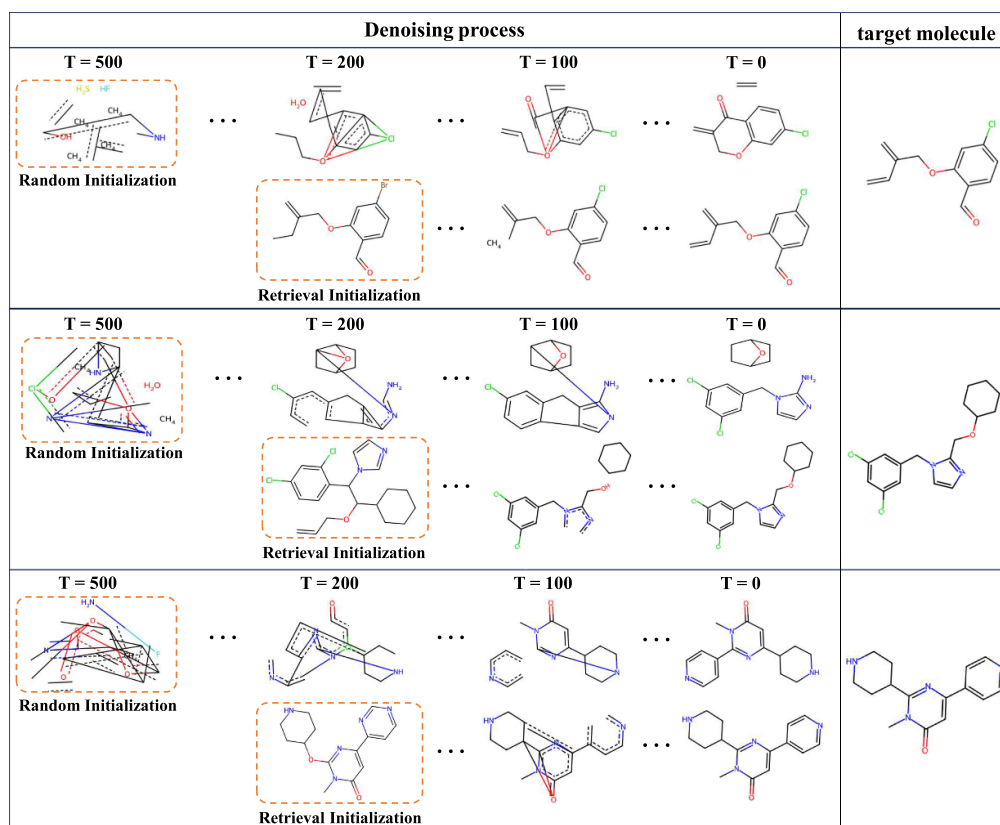


Figure 5. Examples of structure prediction for two target molecules using two inference strategies. (i) Random initialization starts from a random graph and performs 500 denoising steps, resulting in incorrect predictions. (ii) Retrieval Initialization selects a similar molecule from the database based on cosine similarity between NMR-encoded and molecular representations, followed by 200 denoising steps, yielding correct predictions matching the target molecules.

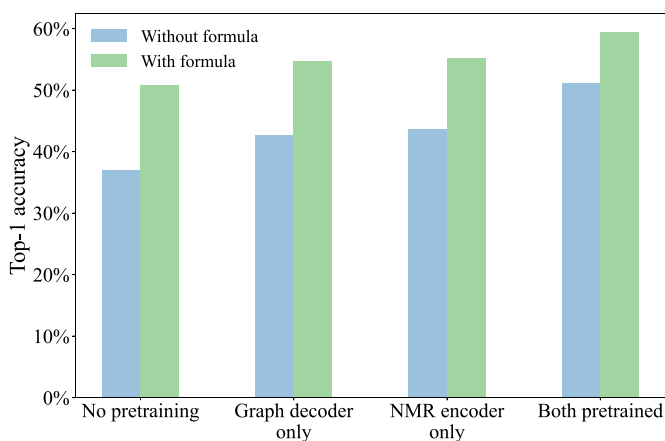


Figure 6. Top-1 accuracy of DiffNMR's molecular structure prediction for molecules with ≤ 15 heavy atoms across different pretraining strategies. Four pretraining scenarios are compared: no pretraining, pretraining only the Diff-AE graph decoder, pretraining only the contrastive learning NMR encoder, and pretraining both components.

5% over the no-pretraining baseline, while pretraining both components yields a synergistic improvement of about 10%. These findings confirm the effectiveness of the proposed two-stage pretraining approach in enhancing the DiffNMR framework's performance.

3. Method

3.1. Notation and problem formulation

We represent a molecule as a graph $G = (V, E)$, where V denotes atoms with attributes in a discrete space $\mathcal{V}$, and E denotes bonds with attributes in $\mathcal{E}$. Node attributes are encoded in a matrix $\mathbf{V} \in \mathbb{R}^{N \times a}$, where N is the number of atoms, $a = \dim(\mathcal{V})$, and each row $\mathbf{v}_i$ is a one-hot vector for atom i . Edge attributes are encoded in a sparse tensor $\mathbf{E} \in \mathbb{R}^{N \times N \times b}$, with $b = \dim(\mathcal{E})$, where $\mathbf{e}_{ij}$ indicates the bond type between atoms i and j . We denote G^t for $t = 1, \dots, T$ as a sequence of noisy graphs where t indicates the index of diffusion steps. NMR interpretation can be formalized as molecular generation under strict spectral constraints, where the goal is to model the true conditional distribution $q(G|y)$ where y denotes conditional signal. To achieve this, we parameterize the distribution as $p_\theta(G|y)$ and implement it using a conditional denoising network $\phi_\theta(G^t, t, y)$, which leverages y to guide the denoising process of a noisy graph aligning the molecule with the NMR.

3.2. Discrete diffusion for conditional graph generation

Diffusion models consist of forward diffusion and reverse denoising processes. The diffusion process $q(G^{1:T}|G^0) = \prod_{t=1}^T q(G^t|G^{t-1})$ corrupts molecular graph $G = G^0 = (V^0, E^0)$

into noisy versions $G^t = (V^t, E^t)$, and as $T \rightarrow \infty$, $q(G^T)$ converges to a stationary distribution $\pi(G)$. The reverse process $p_\theta(G^{0:T}) = q(G^T) \prod_{t=1}^T p_\theta(G^{t-1}|G^t)$, parameterized by θ , reconstructs the graph from the noisy states G^t . The diffusion process perturbs G^0 in a discrete state-space to capture molecular structure. Transition matrices $\mathbf{Q}_V \in \mathbb{R}^{a \times a}$ and $\mathbf{Q}_E \in \mathbb{R}^{b \times b}$ govern the noise for the nodes V and edges E of the graph, respectively. Each diffusion step is defined as $q(G^t|G^{t-1}) = (\mathbf{V}^{t-1}\mathbf{Q}_V^t, \mathbf{E}^{t-1}\mathbf{Q}_E^t)$. As the process is Markovian, the noisy states G^t can be built from G^0 using

$$q(G^t|G^0) = (\mathbf{V}\bar{\mathbf{Q}}_V^t, \mathbf{E}\bar{\mathbf{Q}}_E^t), \quad (1)$$

for $\bar{\mathbf{Q}}_V^t = \mathbf{Q}_V^1 \dots \mathbf{Q}_V^t$ and $\bar{\mathbf{Q}}_E^t = \mathbf{Q}_E^1 \dots \mathbf{Q}_E^t$. Vignac *et al* defined $\pi(G) = (\mathbf{m}_V \in \mathbb{R}^a, \mathbf{m}_E \in \mathbb{R}^b)$ as the marginal distributions of atom types and bond types, and

$$\bar{\mathbf{Q}}^t = \bar{\alpha}^t \mathbf{I} + (1 - \bar{\alpha}^t) \mathbf{1}\mathbf{m}', \quad (2)$$

for atoms or bonds, where $\mathbf{m}'$ denotes the transposed row vector. The cosine schedule is typically chosen for $\bar{\alpha}^t = \cos(0.5\pi(t/T + s)/(1+s))^2$.

The reverse process, which generates G^0 iteratively conditioned on y starting with an initial sample $G^T \sim \pi(G)$, can be implemented by estimating the distribution on the graph $p_\theta(G^{t-1}|G^t, y)$ using denoising neural network $\phi_\theta(G^t, t, y)$. This distribution can be modeled as a product over nodes and edges:

$$p_\theta(G^{t-1}|G^t, y) = \prod_{1 \leq i \leq N} p_\theta(v_i^{t-1}|G^t, y) \prod_{1 \leq i, j \leq N} p_\theta(e_{ij}^{t-1}|G^t, y) \quad (3)$$

where $p_\theta(v_i^{t-1}|G^t, y)$ and $p_\theta(e_{ij}^{t-1}|G^t, y)$ are respectively marginalized over predictions of node and edge types:

$$\begin{aligned} p_\theta(v_i^{t-1}|G^t, y) &= \sum_{v \in \mathcal{V}} p_\theta(v_i^{t-1}|v_i = v, G^t, y) \\ &\quad \times p_\theta(v_i = v|G^t, y) \\ p_\theta(e_{ij}^{t-1}|G^t, y) &= \sum_{e \in \mathcal{E}} p_\theta(e_{ij}^{t-1}|e_{ij} = e, G^t, y) \\ &\quad \times p_\theta(e_{ij} = e|G^t, y) \end{aligned} \quad (4)$$

in which $p_\theta(v_i^{t-1}|v_i = v, G^t, y) = q(v_i^{t-1}|v_i = v)$ and $p_\theta(e_{ij}^{t-1}|e_{ij} = e, G^t, y) = q(e_{ij}^{t-1}|e_{ij} = e)$.

The denoising neural network $\phi_\theta(G^t, t, y)$ with trainable parameters θ takes both noisy graph $G^t = (\mathbf{V}^t, \mathbf{E}^t)$ and condition y as inputs, and aims to predict the clean graph G^0 . This neural network ϕ_θ could be trained to minimize the cross-entropy loss $\mathcal{L}$ between the predicted probabilities $p_\theta(G^0|G^t, y)$ and the true graph G^0 :

$$\begin{aligned} \mathcal{L}(G^0, p_\theta(G^0|G^t, y)) &= \sum_{1 \leq i \leq N} \text{CE}(\mathbf{v}_i, p_\theta(v_i|G^t, y)) \\ &\quad + \lambda \sum_{1 \leq i, j \leq N} \text{CE}(\mathbf{e}_{ij}, p_\theta(e_{ij}|G^t, y)), \end{aligned} \quad (5)$$

where λ determines the weighting between the contributions of nodes and edges.

3.3. Contrastive learning

In the second pretraining stage, a contrastive learning strategy is utilized to train an NMR encoder, aligning NMR spectra with molecular representations, so that the generated NMR spectral feature vectors closely correspond to the molecular feature space established in the first stage. This is accomplished by designating matching spectrum-molecule pairs as positive and non-matching pairs as negative. The NMR encoder is optimized to reduce the difference between feature vectors of positive pairs while increasing the difference for negative pairs. During this process, the molecular encoder's parameters remain unchanged, allowing the NMR encoder to adapt to the existing molecular representation framework.

The training is driven by a contrastive loss function InfoNCE, defined as:

$$\mathcal{L}_{\text{contrastive}} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\text{score}(\mathbf{z}_i, \mathbf{m}_i)/\tau)}{\sum_{j=1}^N \exp(\text{score}(\mathbf{z}_i, \mathbf{m}_j)/\tau)} \quad (6)$$

Here, N represents the batch size, $\mathbf{z}_i$ is the feature vector from the i th NMR spectrum extracted by the NMR encoder, $\mathbf{m}_i$ is the corresponding molecular feature vector from the fixed molecular encoder, $\text{score}(\cdot, \cdot)$ denotes the similarity metric, and τ is the temperature hyperparameter.

The similarity between vectors is computed using a dot-product-based score:

$$\text{score}(\mathbf{z}_i, \mathbf{m}_j) = \frac{\mathbf{z}_i^\top \mathbf{m}_j}{\sqrt{|\mathbf{z}_i|^2 |\mathbf{m}_j|^2}} \quad (7)$$

This score prioritizes directional alignment, enabling the NMR encoder to capture semantic relationships between spectra and molecules effectively. Through this contrastive learning approach, the NMR encoder generates spectral representations that are both meaningful and compatible with molecular features, facilitating integration with the diffusion decoder and enabling efficient spectrum-to-molecule retrieval during inference.

4. Conclusion

In this work, we introduce DiffNMR, an end-to-end framework for molecular structure elucidation from NMR spectra. By leveraging a conditional discrete diffusion model, DiffNMR constructs molecular graphs through holistic, iterative refinement, which enforces global structural consistency and mitigates error cascades. Across benchmarks, DiffNMR significantly outperforms autoregressive baselines such as LL-NMR in domain generalization and robustness—a critical advantage in real-world settings where experimental data are rarely pristine. The framework's effectiveness is further enhanced by a specialized two-stage pretraining strategy and advanced inference techniques. We show that similarity-based filtering, enabled by a well-aligned latent space learned

via contrastive pretraining, substantially improves prediction accuracy. Retrieval initialization further improves performance, mitigates the accuracy drop often observed as molecular complexity increases, and accelerates sampling. Even when an exact match is unavailable, DiffNMR proposes structurally relevant candidates with high similarity to the target molecule, making it a practical tool for chemists to generate and refine hypotheses.

5. Future perspectives

While this work utilizes DiffNMR to demonstrate the potential of generative models in spectral analysis, it also highlights key challenges that the broader field must address to achieve practical utility. A primary hurdle for current AI-driven approaches is the gap between synthetic training environments and experimental realities. Future research in this domain will likely focus on improving robustness against the solvent effects, impurities, and phase distortions common in authentic laboratory data, potentially through advanced domain adaptation or transfer learning techniques.

Furthermore, the evolution of structure elucidation tools necessitates moving beyond 1D spectral data. Just as chemists rely on correlations to solve complex puzzles, computational models must evolve to integrate 2D NMR experiments (e.g. HSQC, HMBC). This multimodal integration is not merely an additive feature but a fundamental requirement to resolve the structural ambiguities that limit current 1D-based methods. Finally, expanding the capability of these models to handle larger natural products (beyond the current heavy atom limitations) will be essential for their deployment in pharmaceutical research and complex mixture analysis.

Code availability statement

This model is developed on the PaddlePaddle which is parallel distribution deep learning framework, and PaddleMaterial which is data-driven and mechanism-driven toolkit for materials R&D. The code of DiffNMR is available at <https://github.com/PaddlePaddle/PaddleMaterial>.

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Conflicts of interest

There are no conflicts to declare.

Author's contributions

Q.Y.: Methodology, Software, Data curation, Formal analysis, Visualization. B.W.: Conceptualization, Methodology, Software, Investigation, Writing—original draft. X.L.: Software, Data curation, Visualization. B.C.: Validation, Formal analysis. W.L.: Data curation, Validation. G.L.: Writing—review & editing, Supervision. X.C.: Writing—review & editing, Supervision, Funding acquisition. M.X.: Writing—review & editing, Supervision. All authors have read and agreed to the published version of the manuscript.

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